

Question1

A researcher is analyzing the therapeutic response of a BET inhibitor in diffuse large B-cell lymphoma (DLBCL). There are 10 Drug Sensitive samples(S1-S10) and 10 Drug Resistant samples(R1-R10). The researcher measured the RNA expression levels of 12 candidate genes, the expression levels are normalized and log2-transformed.

See data file: "homework4-Q1-data.csv".

- (1) Establish an automated statistical analysis pipeline. For each gene, perform a Shapiro–Wilk normality test and a homogeneity of variance test, and automatically select an appropriate hypothesis testing method based on the test results (e.g., Student's t-test, Welch's t-test, or Wilcoxon rank-sum test). In the output data frame, report the gene name, the hypothesis testing method used, the differential expression p-value, and the log2 fold change (log2FC), where $\log_2FC = \log_2(\text{avg of condition B} / \text{avg of condition A})$
- (2) What is the formula for the BH-FDR correction? How should this formula be interpreted? Apply the BH-FDR method to perform multiple testing correction on the statistical results obtained in the previous step.
- (3) For samples involving m independent hypothesis tests, the Bonferroni correction formula is: $p_{\text{adj}} = p \times m$. This is a more stringent correction method. In practical analyses, RNA-seq studies tend to prefer BH-FDR over Bonferroni correction. Please explain the possible reasons for this preference.
- (4) In the hypothesis testing workflow, why is it necessary to perform normality tests and homogeneity of variance tests before conducting a t-test? In what ways can outliers in the data affect the results of a t-test?

Question2

A biology research team is investigating whether a new plant growth hormone can promote the growth of rice seedlings.

The researchers randomly divide the rice seedlings into two groups:

- Experimental group: treated with the new plant growth hormone
- Control group: treated with a standard nutrient solution

After two weeks, the height of the seedlings (cm) is measured. The significance level is set at $\alpha = 0.05$, and the researchers aim to detect a medium effect size (Cohen's d) of 0.6.

1. If each group contains 35 rice seedlings, what is the statistical power of the study?
2. If the researchers want the statistical power to reach 80%, what is the minimum

sample size required for each group?

3. Due to limited greenhouse space, only 20 rice seedlings can be accommodated in each group. Calculate the minimum detectable effect size for the experiment.

Hint: using `pwr.t.test()`

Question3

A researcher is investigating the susceptibility of four different cell lines—CHO, HEK293T, Vero, and BHK-21—to a recombinant GFP-expressing virus. The infection efficiency is quantified as the percentage of GFP-positive cells observed under fluorescence microscopy 48 hours post-infection. Ten independent replicate ROIs were used for each cell line (see “homework4-Q3-data.csv”).

- (1). Perform a step-by-step one-way ANOVA at the 5% significance level to determine whether there is a significant difference in the mean infection efficiency among the four cell lines.
- (2). If the ANOVA results are significant, conduct a Tukey’s HSD post-hoc test to identify which specific cell line differs significantly in their susceptibility to the virus and plot a boxplot of the infection efficiency data. (Hint: Use `TukeyHSD()` in R)

Question4

In a biological experiment investigating the influence of varied housing conditions on mouse body weight, 60 eight-week-old mice with uniform initial weights were randomly allocated into six distinct groups. These groups were maintained for two weeks under specific rearing conditions defined by combinations of:

Factor A (Housing Temperature): Levels at 4°C (A1), 25°C (A2), and 30°C (A3).

Factor B (Diet): Levels consisting of a standard/normal diet (B1) and a high-fat diet (B2).

See data file: “homework4-Q4-data.csv”

- (1) Check if the body weight data meets the assumptions of normality and homogeneity of variance for Factors A (Temperature) and B (Diet).
- (2) Analyze whether Factor A (Temperature) and Factor B (Diet) have significant effects on body weight, without considering the interaction effect between them.

(3) If the interaction effect is considered, re-analyze the above problem and identify which groups show a significant difference in mean body weight compared to the A2B1 group.

Question5

A pharmacologist wants to compare the effects of three different drug treatments (Drug A, Drug B, Drug C) on weight gain in mice. Because the data do not satisfy the normality assumption and variances are unequal, one-way ANOVA cannot be used. The experiment uses a completely randomized design, with 10 mice fed under each drug treatment. After four weeks, the weight gain (in grams) for each mouse is recorded as follows:

- Drug A: 145, 152, 138, 160, 147, 155, 142, 149, 153, 140
- Drug B: 136, 129, 142, 131, 140, 135, 138, 133, 145, 130
- Drug C: 120, 125, 118, 132, 121, 128, 115, 124, 130, 122

Determine whether there is a significant difference in weight gain among the three drug treatments at a significance level of $\alpha = 0.05$. (use R)

